Unit 6 Pre-Test – Lines and Planes Unit 6

MCV4U

Jensen

1) Write a vector equation for the line given its direction vector $\vec{m} = [3,1]$ and point on the line $P(2,7)$.

2) Write the vector equation of the line through each pair of points.

a) $A(1,7)$ and $B(4,10)$

b) $A(-3,5)$ and $B(-2,-8)$

3) Determine if the point $P(-1,11)$ is on the line $[x,y] = [3,1] + t[-2,5]$

4) Write the parametric equation of each vector equation

a) $[x, y] = [10,6] + t[13,1]$

b) $[x, y, z] = [3,0, -1] + t[6, -9,1]$

5) Write the vector equation of each parametric equation

a) $\ell: \begin{cases} x = 1 + 4t \\ y = -6 + t \\ z = 2 - 2t \end{cases}$

b) $\ell: \begin{cases} x = 7 \\ y = -t \\ z = 0 \end{cases}$

6) Given the points $A(3,4, -5)$ and $B(9, -2,7)$, write a vector equation and the parametric equations of the line through A and B.

7) Determine if $\ell_1: \begin{cases} x = 3 - 2t \\ y = 4 + t \end{cases}$ and $\ell_2: 2x + 4y - 7 = 0$ are parallel, perpendicular, or neither. Explain.

8) Determine a vector equation of each line.

a) perpendicular to line $4x - 3y = 17$ and through point $P(-2,4)$

b) parallel to the z-axis and through point $P(1,5,10)$

c) parallel to $[x, y, z] = [3,3,0] + t[3, -5, -9]$ with an x-intercept of -10

d) with the same x-intercept as $[x, y, z] = [3,0,0] + t[4, -4,1]$ and the same z-intercept as $[x, y, z] = [6, -2, -3] + t[3, -1, -2]$

9) Does the point $P(1,2,0)$ lie on the plane $4x + 3y - 5z = 10$?

10) Find the x, y, and z-intercepts of each plane

a) $4x + 2y - 7z + 14 = 0$

b) $3x + 6z + 18 = 0$

11) Write the parametric equation of the plane $[x, y, z] = [0, -4,1] + s[1,10, -1] + t[0,3,4]$

12) Write the vector equation of the plane π: $\begin{cases} x = 2 + s + 7t \\ y = 12s - 8t \\ z = 11 + 6s \end{cases}$

13) Determine if the point $P(-4,-13,10)$ is on the plane $[x,y,z] = [6,-7,10] + s[1,3,-1] + t[2,-2,1]$

14) Determine the x, y, and z-intercepts of $[x,y,z] = [1,8,6] + s[1,-12,-12] + t[2,4,-3]$

15) Write a vector and parametric equation of each plane

a) contains the points $A(1,3,-2)$, $B(3,-9,7)$, and $C(4,-4,5)$

b) has x-int 8, y-int -3, and z-int 2

16) Determine an equation for each plane

a) parallel to the xz-plane; and through the point $A(3,1,5)$

b) containing the points $A(2,1,1)$, $B(5,-3,2)$, and $C(0,-1,4)$

17) Write the scalar equation of a plane given its normal $\vec{n} = [3,7,1]$ and a point $P(3,-6,4)$ on the plane.

18) Write a scalar equation of each plane, given its vector or parametric equation.

a) $[x, y, z] = [3,7,-5] + s[1,2,-1] + t[1,-2,3]$

b) $\pi: \begin{cases} x = 3 - 2s + 2t \\ y = 1 + 3s + t \\ z = 5 - s - 2t \end{cases}$

19) Write a vector and scalar equation of each plane

a) perpendicular to the line $[x, y, z] = [2,4,-9] + t[3,5,-3]$ and including the point $P(4,-2,7)$

b) parallel to the yz-plane and including the point $P(-1,-2,5)$

c) parallel to the plane $3x - 9y + z - 12 = 0$ and including the point $P(-3,7,1)$

d) containing the lines $[x,y,z] = [-2,3,12] + s[-2,1,5]$ and $[x,y,z] = [1,-4,4] + t[-6,3,15]$

20) Determine the angle between the planes $\pi_1: 3x + 2y + 5z = 5$ and $\pi_2: 4x - 3y - 2z = 2$

21) State the intersections of the following systems. Always try to predict the intersection before you spend time solving.

a) $\ell_1: [x, y] = [-12, -7] + s[8, -5]$
$\ell_2: [x, y] = [2, -1] + t[3, -2]$

b) $\ell_1: [x, y, z] = [1, -1, 0] + s[2, -1, 3]$
$\ell_2: [x, y, z] = [1, 2, 3] + t[3, 2, 4]$

c) $\ell_1: [x, y, z] = [7,2,-6] + s[2,1,-3]$
$\ell_2: [x, y, z] = [3,9,13] + t[1,5,5]$

d) $[x, y, z] = [3,1,2] + t[1,-4,-8]$
$4x + 2y - z - 8 = 0$

e) $\ell: \begin{cases} x = 2 + t \\ y = 2 + 2t \\ z = 9 + 8t \end{cases}$

$\pi: 2x - 5y + z - 6 = 0$

f) $\ell: [x, y,] = [3, -2,1] + t[14, -5, -3]$
$\pi: x + y + 3z - 4 = 0$

g) $\pi_1: x + 3y - 4z = 10$
$\pi_2: 3x + 9y - 12z = -30$

h) $\pi_1: 5x - 4y + z = 3$
$\pi_2: x + 3y = 9$

i) $\pi_1: x - 5y + 2z - 10 = 0$
$\pi_2: x + 7y - 2z + 6 = 0$
$\pi_3: 8x + 5y + z - 20 = 0$

j) $\pi_1: 2x + y + 6z - 7 = 0$
$\pi_2: 3x + 4y + 3z + 8 = 0$
$\pi_3: x - 2y - 4z - 9 = 0$

k) $\pi_1: x - y + 2z = 0$
$\pi_2: x - y + 2z + 7 = 0$
$\pi_3: z - 5 = 0$

l) $\pi_1: x + 3y - z = -10$
$\pi_2: 2x + y + z = 8$
$\pi_3: x - 2y + 2z = -4$

22) Find the distance between the point $P(3, -2, 0)$ and the plane $4x - y + 8z = 2$.

23) Find the distance between these skewed lines.

$\ell_1: [x, y, z] = [1, 0, -1] + s[2, 3, -4]$

$\ell_2: [x, y, z] = [8, 1, 3] + t[4, -5, 1]$

1) $[x, y] = [2,7] + t[3,1]$
2)a) $[x, y] = [4,10] + t[1,1]$ **b)** $[x, y] = [-3,5] + t[1, -13]$
3) Yes
4)a) $\ell: \begin{cases} x = 10 + 13t \\ y = 6 + t \end{cases}$ **b)** $\ell: \begin{cases} x = 3 + 6t \\ y = -9t \\ z = -1 + t \end{cases}$

5)a) $[x, y, z] = [1, -6,2] + t[4,1, -2]$ **b)** $[x, y, z] = [7,0,0] + t[0, -1,0]$

6) $[x, y, z] = [3,4, -5] + t[1, -1,2]$; $\ell: \begin{cases} x = 3 + t \\ y = 4 - t \\ z = -5 + 2t \end{cases}$

7) Parallel
8)a) $[x, y] = [-2,4] + t[4, -3]$ **b)** $[x, y, z] = [1,5,10] + t[0,0,1]$ **c)** $[x, y, z] = [-10,0,0] + t[3, -5, -9]$
d) $[x, y, z] = [3,0,0] + t[3,0, -1]$
9) Yes
10)a) x-int: $-\frac{7}{2}$; y-int: -7; z-int: 2 **b)** x-int: -6; y-int: none; z-int: -3

11) $\pi: \begin{cases} x = s \\ y = -4 + 10s + 3t \\ z = 1 - s + 4t \end{cases}$

12) $[x, y, z] = [2,0,11] + s[1,12,6] + t[7, -8,0]$

13) No
14) x-int: 1; y-int: -4; z-int: 3
15)a) $[x, y, z] = [1,3, -2] + s[2, -12,9] + t[3, -7,7]$; $\pi: \begin{cases} x = 1 + 2s + 3t \\ y = 3 - 12s - 7t \\ z = -2 + 9s + 7t \end{cases}$

b) $[x, y, z] = [8,0,0] + s[8,3,0] + t[4,0, -1]$; $\pi: \begin{cases} x = 8 + 8s + 4t \\ y = 3s \\ z = -t \end{cases}$

16)a) $y = 1$ **b)** $[x, y, z] = [2,1,1] + s[3, -4,1] + t[5, -2, -2]$
17) $3x + 7y + z + 29 = 0$
18)a) $x - y - z - 1 = 0$ **b)** $5x + 6y + 8z - 61 = 0$
19)a) $[x, y, z] = [4, -2,7] + s[1,0,1] + t[0,3,5]$; $3x + 5y - 3z + 19 = 0$
b) $[x, y, z] = [-1, -2,5] + s[0,0,1] + t[0,1,0]$; $x = -1$ or $x + 1 = 0$
c) $[x, y, z] = [-3,7,1] + s[3,1,0] + t[1,0, -3]$; $3x - 9y + z + 71 = 0$
d) $[x, y, z] = [-2,3,12] + s[-2,1,5] + t[-3,7,8]$; $27x - y + 11z - 75 = 0$
20) $96.9°$
21)a) $(356, -237)$ **b)** no intersection; skew lines **c)** $(1, -1,3)$ d) $(2,5,10)$
e) no intersection ; parallel and distinct **f)** infinite solutions; parallel and coincident
g) no intersection; parallel and distinct **h)** $[x, y, z] = [9,0, -42] + t[-3,1,19]$
i) $[x, y, z] = [0,2,10] + t[1, -1, -3]$ **j)** $(3, -5,1)$ **k)** inconsistent system; no solution
l) inconsistent system; no solution
22) $\frac{4}{3}$
23) 6.79